

RE-XTRACT: Unanimity-Gated Document Extraction with Calibrated Abstention

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Abstract

In information extraction from unstructured documents, errors have asymmetric costs: a wrong value silently corrupts downstream databases. Although vision-language models (VLMs) are capable of reading document contents with high proficiency, they cannot reliably flag their own mistakes. We introduce RE-XTRACT, an extraction system designed for Indian real-estate regulatory (RERA) filings. To prevent silent failures, RE-XTRACT classifies documents and extracts a fixed schema of dates and building-scope fields using three composable gates over an open-weight VLM: (1) exact-match consistency across N voting rounds, (2) independent validation of unanimous consensus by a frontier model, and (3) a final fallback that safely abstains when they disagree. On a 143-document production benchmark, RE-XTRACT classifies documents with 95.1% accuracy against 85.3% for a previous generator-verifier pipeline, and reads 95.6% of 113 gold dates correctly against 88.5%. We demonstrate RE-XTRACT live, letting users process documents and inspect decision provenance in real time.

Demo —

<https://taxonomic-unadversely-zonia.ngrok-free.dev>

Introduction

Whether a real-estate project in India is on schedule, delayed or delivered is largely invisible to the homebuyers and investors who depend on it. The Real Estate (Regulation and Development) Act, 2016 (RERA) addresses this trust gap by requiring every real-estate project in India to register with its state regulatory authority and to keep a public document trail. In practice this trail is a directory of scanned PDFs per project, uploaded by promoters and authorities over the project’s lifetime. Four document types carry the information that downstream products actually need:

- **Registration Certificate (RC)** - establishes the project and states the date on which registration was granted and the date by which the promoter has promised completion.
- **Extension Certificate (EC)** - issued later, moves the promised completion date in cases of delay.
- **Occupancy Certificate (OC)** - a municipal body certifies a building is fit for occupation.
- **Completion Certificate (CC)** - a municipal body certifies construction is complete and conforms to plan.

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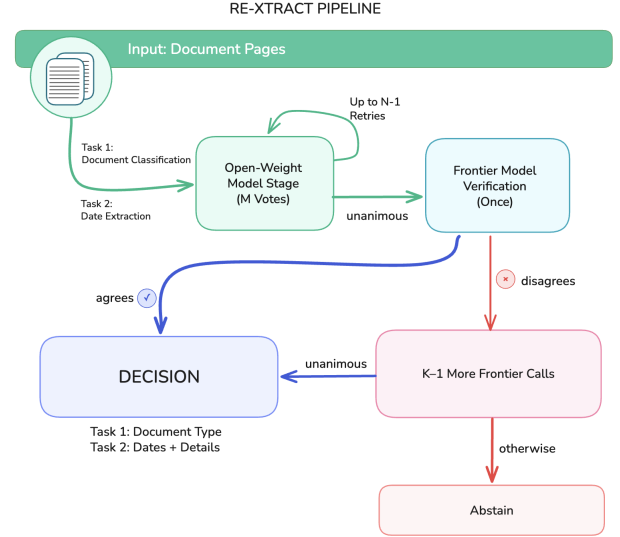


Figure 1: The RE-XTRACT pipeline. A round of M votes from one open-weight model must be unanimous, with up to $N-1$ retries, before a frontier model checks the answer. A rejected answer escalates to K frontier calls in total, and anything still unresolved abstains.

Altogether, these dates are the ground truth for a listed project’s current and future status; extracting them, however, is not trivial. A single project’s filings run to hundreds of documents, most of which - GST registrations, Water certificates, handover deeds - carry none of these dates. Moreover, the relevant certificates follow no common format, and some are handwritten.

Several works explore Optical Character Recognition (OCR) technique for extraction of fields from documents (Zhang et al. 2024), but the nature of unstructured documents renders them unsuitable. Instead, vision-language models (VLMs) have gained traction, since they read a page directly as an image and thus absorb layout variation that a fixed OCR pipeline cannot (Han et al. 2025). Additionally, established OCR toolkits are themselves converging on this design: PaddleOCR-VL folds a 0.9B-parameter VLM into the parsing stack in place of a staged recogniser, which enables it to parse across 109 languages (Cui et al. 2025).

A single VLM call, however, yields no signal that separates a correct reading from an incorrect one. A date transcribed from the wrong document is as fluent as the true value. The two errors are not equally costly in this setting; an omitted date can be re-queued, but an erroneous one is republished as the project’s official status.

This asymmetry shapes RE-XTRACT. The system follows the principle of *Agreement-as-a-Confidence* by sampling one model repeatedly and requiring exact agreement before corroborating it across a second model, and abstaining when neither holds. Sampling consistency is a known fabrication signal (Wang et al. 2023; Manakul, Liusie, and Gales 2023); RE-XTRACT composes it with cross-model corroboration with an explicit reject option (Geifman and El-Yaniv 2017; Kamath, Jia, and Liang 2020), and preserves the entire decision history for auditable tracking.

System Overview

RE-XTRACT takes a RERA ID and collects the project’s filed documents. Figure 1 shows the pipeline, a fixed Lang-Graph state machine. Each document is ran in two stages - it is first classified into one of the four certificate types, and then routed to its own schema of dates and details. Both decisions pass through the same three gates below.

1. Ingest: Documents are rendered to page images at 150 DPI and deduplicated using MD5 for byte-identical PDFs, and perceptual hash for visually identical rescans.

2. Voting: The pages are put to a self-hosted open-weight VLM M times, giving M votes on the decision at hand. On the extraction pass, dates are copied verbatim and converted to ISO form (YYYY-MM-DD) by a deterministic parser, so votes are compared on value rather than wording. The votes are then checked for a unanimous match; if they disagree, the round is retried up to $N-1$ times, lasting at most N rounds.

3. Verification: Unanimity among votes from one model is not enough on its own, since a small model can be consistently wrong. When a round is unanimous, RE-XTRACT puts the same question to a frontier model over the same pages. If its answer matches the unanimous one, the answer is final.

4. Escalation and Abstention: If the frontier model disagrees, RE-XTRACT calls it $K-1$ more times and checks all K answers for unanimity. If they agree, that answer is final. Otherwise RE-XTRACT abstains: a classification decision sets the document aside, an extraction decision writes no value, and either is terminal. K is capped separately from M because frontier calls are metered and local votes are not.

If any of the gates abstain from answering, the document is skipped, or flagged for manual review.

Demonstration

Our demo shows the full extraction workflow on a fixed corpus of RERA filings. A visitor selects one of six preloaded projects and the pipeline runs on it in-real time. The interface shows each of the M votes as the open-weight model is called accompanied by the frontier model’s verdict, and subsequent

Class	Gen.-verifier			RE-XTRACT		
	P	R	F1	P	R	F1
<i>Stage 1: Document Classification</i>						
Registration	.98	.96	.97	1.00	.98	.99
Extension	.80	1.00	.89	1.00	1.00	1.00
OC / Completion	.78	1.00	.87	.97	.94	.95
Out-of-schema	1.00	.32	.49	.84	.93	.88
Accuracy	85.3%			95.1%		
Macro-F1	.804			.955		
<i>Stage 2: Date Extraction, 113 gold dates</i>						
Correct	100			108		
Incorrect	13			5		
Accuracy	88.5%			95.6%		

Table 1: Both gated stages, compared against the earlier generator-verifier pipeline. Classification covers the 143-document benchmark (45 Registration, 8 Extension, 62 Occupancy/Completion, 28 out-of-schema); date extraction covers 113 gold dates from the same projects.

flows upon disagreement. The complete process is shown in the demo video.

Deployment and Results

RE-XTRACT runs in production at Housing.com, writing RERA filings into a Delta Lake table behind consumer-facing project timelines. The local model is a 2.3B-parameter model from the Qwen 3.5 family (Bai et al. 2023); the frontier model is gemini-2.5-flash-lite (Comanici et al. 2025). We evaluate classification on 143 documents from production traffic, covering 47 projects across seven states.

Table 1 compares RE-XTRACT against the generator-verifier loop (Madaan et al. 2023), in which one call extracted and a second call verified until the two agreed. Classification accuracy rises from 85.3% to 95.1% and Macro-F1 score from .804 to .955. Biggest improvement is in classification of out-of-schema documents, where the recall moves from .32 to .93, so the documents that should have been turned away are now correctly flagged. Precision on this class, however, drops from 1.00 to .84, which is an acceptable tradeoff, since this is an error that costs an operator a second look rather than a corrupted record.

The lower panel of Table 1 reports the date extraction stage. Over 113 gold dates from the same projects, RE-XTRACT reads 108 correctly against 100 for the generator-verifier loop, raising date accuracy from 88.5% to 95.6%.

Conclusion

We present RE-XTRACT, that treats agreement as its only evidence of correctness: a local model must vote unanimously, a frontier model must confirm it, and anything unresolved is set aside, not guessed. In production, that is 95.1% classification accuracy and 95.6% on dates.

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